Departement Elektriese, Elektroniese en Rekenaar-Ingenieurswese	Department of Electrical, Electronic and Computer Engineering

Eksamen / Examination

Module EBN111

Elektrisiteit en Elektronika / Electricity and Electronics

4 Junie 2010 / 4 June 2010

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting: ***Exam information:***			
Maksimum punte: *Maximum marks:*	90	Volpunte: *Full marks:*	90
Duur van vraestel: *Duration of paper:*	180 minute *180 minutes*	Oopboek / toeboek: *Open / closed book:*	Toe *Closed*

Memo

Dosent(e): *Lecturer(s):*	1 J Joubert		
	2 D Bhatt		
	3 B Mabuza		
Groephoof: Ek bevestig dat die vraestel die uitkomstes toets soos gespesifiseer in die studiehandleiding. *Group head: I confirm that the question paper evaluates the outcomes as specified in the study guide.*			
Groephoof: *Group Head:*	Prof J Joubert	Handtekening: *Signature:*	

EBN111 EKSAMEN ANTWOORDBLAD/ EXAM ANSWER SHEET
4 JUNIE/JUNE 2010

Student se besonderhede/*Student's details*										
Van en voorletters: *Surname and initials:*		Studentenommer: *Student number:*								
Graadkursus: *Degree Course:*										
Tel. nr.: *Tel no.:*										

PUNTE/MARKS									
Q1	/30	**Q2**	/25	**Q3**	/15	**Q4**	/20	**Final**	/90

VRAAG /QUESTION 1 [30]		
1.1	$W = 270MJ$	$N_{bat} = 10$
1.2	$Q = 2160C$	$P = 0.292W$
1.3	$P_V = -5W$	$P_I = -40W$
1.4	$G_{eq} = 1S$	
1.5	$I_3 = 12mA$	$V_2 = 15V$
	$I_3 = 12mA$	
1.6	$a = 3$	$b = 10$
	$c = 6$	$d = 60$
1.7	$e = 2$	$f = 15$
	$g = 12$	$h = 50$
1.8	$\Delta = 14$	$v_2 = 1.43V$
1.9	$I_B = 40.6\mu A$	$V_{CE} = 12.475V$
1.10	$I = 7.5\mu A$	$v_0 = -15V$

VRAAG /QUESTION 2 [25]		
2.1	$v_o = 40V$	
2.2	$V_s = 50V$	$R_s = 300\Omega$
2.3	$v_{1(10V)} = -1.43V$	$i_{1(0.5A)} = -0.18A$
2.4	$R_L = 3.33K\ \Omega$	$P_{max} = 83.4MA$
2.5	$R_{TH} = 6.55\ \Omega$	
2.6	$V_{TH} = 312V$	$R_{TH} = 72\ \Omega$
2.7	$R_{TH} = 66.7\ \Omega$	
2.8	$I_N = 7.5mA$	$R_N = 85.7\ \Omega$
2.9	$R_N = 0\ \Omega$	

VRAAG 3 – QUESTION 3 [15]			
3.1	$v_o =$ *8mV*	$i_o =$ $-2\mu A$	
3.2	$V_{out} = -15V$		
3.3	$V_1 = -60mV$	$V_2 = 0.5mV$	$V_3 = 602.75mV$
3.4	$L_{eq} = 11.5H$		
3.5	$C_{eq} = 35\mu F$		
3.6	$W_C = 400nJ$	$W_L = 0.18J$	$W_{10\Omega} = 5.76Kj$ OR *1.6Wh*

VRAAG 4 – QUESTION 4 [20]		
4.1	(a) $v(t) =$ 24.8 Cos (ωt + 36^o) V	
	(b) $\overline{V}_1$, 160^o	
	(c) $\overline{V}_2 =$ 0+j2 V	
4.2	(a) 0.38 +j0.87	
	(b) 3.76 $\angle$50.8^o	
4.3	$\overline{a}$ = 2 + j5	$\overline{b}$ = -j5
	$\overline{c}$ = -j5	$\overline{d}$ = 1 + j3
4.4	(a) $i(t)$ = 2.06 Cos (400t – 36.7^o) A	
	(b) Z_{ab} =1.47 $\angle$ 79.4^o Ω	
4.5	$\overline{V}_{th}$ = 8 $\angle$173.13^o V	